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Impact of AI-generated virtual streamer interaction on consumer purchase intention: A focus on social presence and perceived value

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ABSTRACT

The immense growth of virtual streaming is reshaping shopping habits. However, limited research, if any, has focused on the differentiated effects of interaction types—namely product interactions versus social interactions—used by virtual streamers, and their distinct impacts on purchase intentions. This study, focusing on China's booming virtual streamer market, examines how artificial intelligence (AI)-generated virtual streamers' interaction types impact consumer purchase intention, considering the mediating roles of social presence and perceived value, and the moderating effect of product types. Utilizing a randomized scenario experiment among young consumers, the data was collected during a cross-sectional timeframe for two case scenarios, from which we analyzed 100 valid responses in Case 1 and 201 valid responses in Case 2. Results showed that product-focused interactions enhance purchase intention for utilitarian products via perceived value, while social interactions boost purchase intention for hedonic products through social presence. These findings suggest that AI-generated virtual streamers can strategically tailor interaction styles based on product categories to maximize livestream commerce effectiveness. This study enriches understanding of digital consumer behavior and provides actionable insights for retailers deploying AI-generated virtual streamers. It also highlights social presence and perceived value as key psychological mechanisms in shaping consumer responses to virtual streamer interactions.

1. Introduction

With advancements in 5G technology, artificial intelligence (AI), virtual reality, and augmented reality, live commerce is reshaping consumer shopping habits (Bharadwaj et al., 2022). Among these technological innovations, virtual streamers, highly realistic digital personas, are assuming pivotal roles by leveraging their 24/7 availability, cost-effectiveness, and interactive content—especially excelling at night-time broadcasts (Wu et al., 2023). A report on China's Virtual Anchor Industry forecasts the market to reach 270 billion RMB by 2030 (iiMedia Research, 2023), driven by advantages like uninterrupted services, cost savings, and minimized risks compared to human streamers (Hu and Ma, 2023; Mouritzen et al., 2024). Kuaishou, a leading short video and livestreaming platform in China, hosted the virtual streamer "Guan Xiaofang" with significant success, garnering over a million views and 172 thousand likes upon debut, later surging to 2.3 million views and 1.2 million likes (iiMedia Research, 2023). Similarly, during JD. com's '6.18' shopping festival, virtual streamers conducted over 380,000 hours of livestream, attracting millions of engaged users, showcasing the significant market appeal and sales potential of AI-driven livestreaming (Hu, 2024).

While virtual streamers excel in brand collaborations, live sales, and social interactions, their market penetration remains below 1 % (Ebrun, 2023; Wu et al., 2023). This highlights challenges in converting virtual streamers' appeal into actual purchases, underscoring the need for enhanced interactive experiences and differentiated strategies to boost purchase intentions (Chen et al., 2024). Existing literature extensively explores live commerce's impact on consumer behavior across multiple dimensions, focusing on streamer characteristics (Guo et al., 2022; Song et al., 2024; Wang et al., 2024), product type compatibility (Yang et al.,

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2023; Zhen et al., 2024), social presence and flow experience (Kim and Kim, 2022; Zheng et al., 2023), and interaction types (J. Chen et al., 2023; Fan et al., 2024; Peng et al., 2024). Most insights, however, are derived from human streaming contexts. Human streamers excel in emotional expression, spontaneity, and trust-building, whereas virtual streamers' strengths lie in precise recommendations, personalization, rapid information processing, and strong product knowledge (Ham et al., 2024; Stein et al., 2024; Thomas and Fowler, 2021). This points to the need to validate existing findings within virtual streaming contexts, particularly regarding how interaction types influence purchase intentions.

The interaction between virtual streamers and consumers can be categorized into two fundamental types of engagement: product interactions and social interactions. Product interactions refer to conveying product-related information. Virtual streamers enhance perceived value and trust through product demonstrations, tailored language, and credible interactivity, boosting engagement and purchase intentions (Akhtar et al., 2024; Wang et al., 2025; Yang et al., 2024). These interactions aim to enhance perceived product value by addressing consumers' cognitive engagement and rational decision-making processes. Social interactions, on the other hand, focus on virtual streamers' ability to establish emotional and relational connections with consumers through personalized communication, emotional expressions, and social-oriented language (Wu et al., 2023; Yao et al., 2024; Yu et al., 2024). Such interactions foster social presence and build para-social relationships, enhancing trust and emotional intimacy with the audience (Srivastava and Chandra, 2018).

Although existing research has examined the general impact of virtual streamers on consumer behavior (Gao et al., 2023), there is limited previous work investigating the specific roles of different interaction types. Some studies have explored the perceived value provided by virtual streamers (Akhtar et al., 2024; Yang et al., 2024; Wang et al., 2025) or their influence on emotional and relational connections (Yao et al., 2024; Wu et al., 2023; Yu et al., 2024), but these studies tended to focus on isolated aspects—they did not differentiate between the roles of product interactions and social interactions or consider how these interaction types, individually or collectively, influence purchase intentions. This lack of differentiation limits both the theoretical and practical understanding of how virtual streamers can strategically balance interaction types to enhance purchase intentions.

Theories such as perceived value theory and social presence theory have been used to explain the mechanisms underlying virtual streamer interactions. However, each theory has its own focus and scope. Perceived value theory primarily emphasizes product features and rational decision-making, which may not fully account for the emotional needs inherent in virtual interactions. In contrast, social presence theory highlights emotional connectivity and social engagement but places less emphasis on the role of product functionality in influencing purchasing decisions (Liao et al., 2023; Yao et al., 2024). These perspectives indicate that neither theory alone captures the full complexity of how virtual streamer interactions affect purchase intentions, suggesting the need for an integrated framework to address these complementary aspects.

To address these gaps, this study focuses on three critical inquiries: (1) Examining how social interactions and product interactions by virtual streamers affect consumer purchase intentions; (2) Elucidating the specific mechanisms through which social presence and perceived value mediate these relationships; (3) Investigating the moderating effects of product types—hedonic and utilitarian—on the aforementioned mediational mechanisms. By addressing these gaps, this study aims to provide an integrated framework that refines the comprehension of consumer decision-making processes in virtual streamer-driven live commerce, offering both theoretical advancements and practical implications. Through these inquiries, this study contributes to the literature by validating the mediating role of social presence and perceived value, providing new empirical evidence for applying these theories in digital interactive environments and enriching their explanatory power in live commerce contexts. In addition, integrating perceived value and social presence theories in an integrated model compensates for the limitations of a single-path approach and provides a comprehensive perspective to understand the influence of virtual streamer interactions on consumer purchase intentions. Incorporating the product type as a moderating variable further enriches the theoretical framework, aligning it closer to practical applications. The integrated framework enables enterprises to leverage virtual streamers to enhance consumer purchase intentions more effectively.

The rest of this paper is structured as follows: Section 2 discusses some background and theoretical frameworks; Section 3 presents hypothesis formulations and the resulting conceptual model; Section 4 covers the methodology, including the research design; Section 5 outlines the findings. Section 6 summarizes the discussion and theoretical and practical implications. Finally, limitations and future research directions are outlined.

2. Theoretical background

2.1. Influence of streamer on consumer behavior in live commerce

The rapid expansion of live commerce underscores the pivotal role of streamers in shaping consumer purchasing decisions. Research has explored this phenomenon across various dimensions, including streamer characteristics, interaction behaviors, emotional expressions, and situational mechanisms, highlighting their profound impact on consumer behavior. Streamer traits such as attractiveness, competence, and humor (Guo et al., 2022; Zhang et al., 2023), dynamic characteristics (Peng et al., 2024), physical and competence-based appeal (Wang et al., 2024), and gender and product assortment depth (Yang et al., 2023), have been shown to significantly influence consumer decisions. Emotional expressions, including positive and negative emotions (Ma et al., 2024) and emotional topics (Luo et al., 2024), enhance consumer engagement and sales, while proactive strategies in failure scenarios (J. Chen et al., 2023) and optimal information density further improve trust, word-of-mouth, and sales performance (Zhang et al., 2023). Interaction behaviors also exert a considerable influence, encompassing consumer-streamer dynamics (Fan et al., 2024), the role of social presence in enhancing purchase intentions (Zheng et al., 2023), and the impact of personalized and responsive interactions in building perceived value and trust (X. Chen et al., 2023). Collectively, these studies provide a comprehensive understanding of the multifaceted ways streamers influence consumer decision-making in live commerce, offering valuable theoretical perspectives and practical insights.

Despite the wealth of theoretical frameworks and empirical evidence explaining the influence of human streamer interactions on consumer behavior, these conclusions may have limitations when applied to virtual streamers. While virtual streamers leverage big data and algorithms to provide highly personalized interactions, excelling in breadth, they confront challenges in delivering depth and emotionally nuanced personalized interactions (Stein et al., 2024). Although proficient in swift information processing and expert knowledge dissemination, fostering a rational purchasing experience (Thomas and Fowler, 2021), virtual streamers falter in emotive expression and improvisational interaction, potentially hindering swift trust establishment and intimacy formation. Despite algorithmically generating entertaining content, virtual streamers lack the authenticity of emotional exchanges (Stein et al., 2024) and the flexibility of spontaneous dialogue characteristic of human streamers (Ham et al., 2024), which might influence emotional resonance and purchase willingness among viewers.

Consequently, there is a pressing need to delve deeper into the mechanisms by which virtual streamer-consumer interactions influence purchasing behavior. Moreover, it is imperative to consider variations in the effectiveness of virtual streamer interactions across different product categories, such as hedonic vs. utilitarian products, to furnish a more comprehensive theoretical framework and practical guidance for live

commerce operations.

2.2. Impacts of virtual streamers on consumer behavior in e-commerce livestreaming

In recent years, a surge of interest has been witnessed in the application of virtual streamers in e-commerce livestreaming. Scholarly endeavors have focused on exploring the implications of virtual streamer-fan interactions on consumer behavior, with a particular emphasis on virtual streamer types, language strategies employed, streamer characteristics, and social interaction modes to consumer purchase intention and brand forgiveness (El Hedhli et al., 2023; Gao et al., 2023; Hu and Ma, 2023; Wu et al., 2023; Xie et al., 2024; Yao et al., 2024). Notably, Xie et al. (2024) explored consumer brand forgiveness of the two types of streamers (real human streamers vs. human-powered virtual streamers) when they make inappropriate remarks. Yao et al. (2024) found that virtual streamers' use of socially oriented language in e-commerce livestreams significantly influences purchase intention under experiential product conditions but less under search product conditions, with perceived warmth and competence as mediating factors. Moreover, Hu and Ma (2023) highlighted that sensory language use by virtual streamers decreases purchase intention unless viewers are aware that the virtual streamer is human-operated, in which case this negative effect can be reversed.

This existing research underscores the multifaceted influence of different interaction modalities virtual streamers employ on consumer behavior, yet several avenues remain uncharted. Specifically, the comprehensive exploration of how distinct interaction types between virtual streamers and consumers affect purchase intention is imperative. Such understanding can empower businesses to craft more effective livestream content, enhancing consumer purchase intention and loyalty. Furthermore, as mediating variables, social presence and perceived value likely play a pivotal role in the relationship between virtual streamer interaction types and purchase intention, fostering a sense of proximity and trust that directly impacts purchasing decisions. The moderating effect of product types (utilitarian vs. hedonic) also deserves deeper scrutiny. Investigating these factors supplements the literature and provides theoretical guidance and practical recommendations for businesses to optimize their livestream strategies, elevate consumer experiences, and ultimately drive higher marketing effectiveness.

3. Hypothesis construction and conceptual framework

Virtual streamers significantly impact consumer purchase intention through interactions, following two pathways: product interactions elevating perceived value and social interactions enhancing social presence. Each path partially explains the influence of virtual streamers' interactions, with the former focusing on information delivery and customization while overlooking social interactions' psychological and emotional impacts and the latter emphasizing emotional connections over product attributes in purchase decisions. However, previous studies have only examined these aspects in isolation. They did not distinguish between the roles of product interactions and social interactions, nor did they examine how these interaction types influence purchase intentions. To address these gaps, the current research integrates both pathways by adopting the Service-Dominant Logic (SDL) framework. SDL emphasizes the co-creation of value between firms and consumers. In this process, the focus should not only be on the product itself but also the customer experience (Vargo and Lusch, 2004). This model suggests that product interaction and social interaction collectively influence consumer purchase intentions through perceived value and social presence. Virtual streamers co-create value with consumers by offering services (information transmission and emotional interaction), which not only enhance the product's perceived value but also strengthen the relationship between consumers and hosts through social presence, thereby increasing overall purchase intentions. This integrated model illustrates how product and social interactions in a virtual environment influence consumer decisions in a service-oriented manner.

3.1. Social presence

Social presence refers to warmth, sociability, personalness, and sensitivity experienced during social interactions (Etemad-Sajadi, 2016). Grounded in social presence theory, when individuals feel each other's presence during interactions, it fosters trust and intimacy, influencing their behaviors and attitudes (Srivastava and Chandra, 2018) Consumers can likewise experience authentic social connections with virtual streamers in virtual environments. Virtual streamers bridge psychological distances by emulating human emotional expressions and spontaneous conversations, augmenting trust and intimacy. This heightened social presence encourages consumers to trust and purchase products endorsed by virtual streamers (Aljaroodi et al., 2022). Using virtual streamers significantly enhances social presence, as Moon et al. (2013) demonstrated that interactions with sales or co-shopper avatars significantly increased social presence. Lu et al. (2016) and Dabiran et al. (2024) affirm the positive role of social aspects in shaping online purchase behaviors. Building upon this, we hypothesize.

H1. *Virtual streamers' social interactions (vs. product interactions) significantly impact consumer purchase intention through the mediating role of social presence.*

3.2. Perceived value

Derived from Zeithaml's (1988) perceived value theory, consumer evaluations of products or services hinge on the balance between perceived utility and costs incurred. When product interactions provided by virtual streamers yield perceived utility exceeding the costs, consumers perceive higher value, leading to increased purchase intention. virtual streamers, leveraging big data and algorithms, deliver personalized interactions, capture user preferences, and offer tailored recommendations, excelling in rapid information processing and expert knowledge conveyance. This elevates consumers' perceived value, enhancing functional appreciation of products and trust in virtual streamers (Jia et al., 2022). Hence, we propose.

H2. *Virtual streamers' product interactions (vs. social interactions) significantly and positively influence consumer purchase intention through the mediation of perceived value.*

3.3. Service-dominant logic and the moderating role of product categories

We introduce the product category (hedonic vs. utilitarian) as a moderator. Different product types steer consumer focus during virtual streamers' interactions. For hedonic products, emotional expression and social presence are prioritized, emphasizing social interactions. Conversely, professional expertise and personalized recommendations take precedence for utilitarian products, highlighting product interactions. Research supports the regulatory role of product categories in virtual streamer-induced consumer purchase intention (Yao et al., 2024). Therefore, we hypothesize.

H3. *For utilitarian products, virtual streamers' product interactions (vs. social interactions) have a more pronounced positive effect on consumer purchase intention through the mediation of perceived value.*

H4. *For hedonic products, virtual streamers' social interactions (vs. product interactions) exert a more significant positive influence on consumer purchase intention through the mediation of social presence.*

Under the SDL framework, the integrated theoretical model (presented in Fig. 1) elucidates how virtual streamers, through personalized recommendations and expert knowledge, enhance perceived value, leading to higher purchase intention, and through emotional

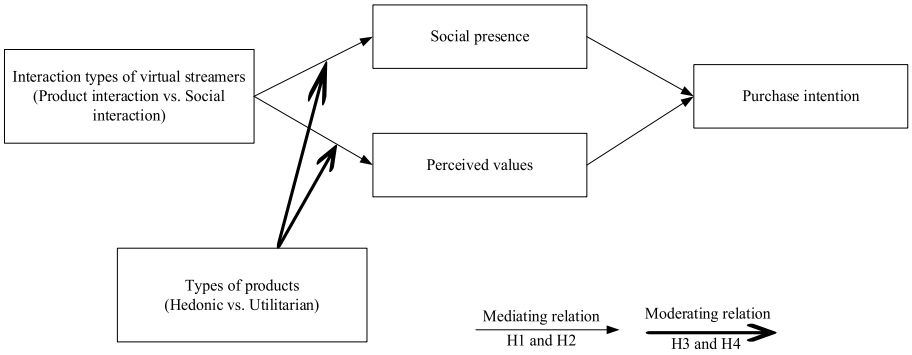


Fig. 1. Conceptual framework.

expressions and spontaneous dialogues, fortify social presence, fostering trust and intimacy, which also increases purchase intention. Product category modulation intensifies either the social or product interaction pathway based on the nature of the product being promoted, providing a comprehensive understanding of virtual streamers' interactions' impact on consumer purchase intention in e-commerce livestreaming.

4. Methodology

In alignment with the proposed theoretical model, this study employed a scenario-based experimental design to test the hypotheses, conducted across Study 1 and Study 2. Study 1 involved two groups of experimental materials and used scenario experiments to validate the mediating effect of social presence and perceived value, addressing H1 and H2. Study 2 included four sets of experimental materials and examined the moderating effects of the type of product, testing H3 and H4.

4.1. Study 1

Study 1 employed situational experiments to examine the mediating effects proposed in H1 and H2, specifically whether virtual streamers' social interactions (vs. product interactions) influence consumer purchase intention through social presence or perceived value.

4.1.1. Pre-study

In December 2023, a pre-study (Pre-study 1a) was conducted to select representative products for the main research. An online



Fig. 2. Video screenshots of Study 1.

questionnaire was distributed to recruit 50 participants from a university in Chengdu, Sichuan. Participants rated ten common products – external hard drives, power banks, perfumes, headphones, shoes, laptops, toothpaste, mobile phones, sunscreen, and watches – on a scale of 1–7. The scale ranged from 1 (purely utilitarian) to 7 (purely hedonic), with 4 representing neutral attributes, based on the products' practicality and pleasure-inducing qualities. The pretest identified watches as a neutral product type, perfume as representative of hedonic products, and external hard drives as illustrative of utilitarian products. The purpose of Study 1 is to examine the mediating effect. To avoid exogenous influences resulting from product types, Study 1 used a watch with neutral "utilitarian vs. hedonic" attributes as the experimental material.

The video and textual materials for this study were primarily derived from livestreams of Taobao and JD.com. By thoroughly examining virtual streamers' discourse and content strategies during livestreams on Taobao and JD.com, this study has developed a virtual brand, A. Queen, and created livestreaming videos that replicate the interactive scenarios typically exhibited product interaction and social interaction by virtual streamers shown in Fig. 2. Drawing from existing research, the study maintains uniformity in the live broadcast environment, including consistent backgrounds, brand imagery, and the virtual streamers' persona. Moreover, to control for variables that might influence the experimental results, the study strictly regulates the video length and the quantity of interactive dialogue, ensuring a balanced approach that mitigates the risk of content bias or other external influences.

To ensure the validity of the experimental materials, another pre-study (Pre-study 1b) was carried out to determine whether the participants could accurately judge the types of streamers in the experimental materials. This pre-study adopted a single-factor (product interaction vs. social interaction) between-subjects design and was conducted on the Chinese online crowdsourcing platform Credamo in December 2023. In the pre-study, 50 participants were recruited online to engage in the experiment, who were randomly assigned to either of the two interaction types (product interaction or social interaction) to view corresponding videos and rate the virtual streamer interaction types using a scale. This scale was adapted from the interaction measurement scale outlined by Yuan and Huang (2022). It was tailored to this study by dividing it into two aspects: product interaction and social interaction, each comprising three measurement items (as shown in Table 1). Results revealed that participants assigned to the social interaction condition had a significantly greater rate of social interaction than those assigned to the product interaction. Specifically, results of a one-way analysis of variance (ANOVA) show that the Social Interaction Material condition was rated significantly higher for social interaction compared to the Product Interaction Material condition ($M_{Social\ interaction} = 5.46$ vs. $M_{product\ interaction} = 2.92$, $F(1,48) = 281.75$, $p < 0.001$). Meanwhile, participants assigned to the product interaction condition had a significantly greater rate of product interaction than those assigned to the social interaction condition ($M_{Social\ interaction} = 5.54$ vs. $M_{Product\ interaction} = 2.54$, $F(1,48) = 320.31$, $p < 0.001$). Therefore, intreraction was effectively manipulated.

Table 1
Measurement items for product interaction and social interaction.

Constructs	Measurement Items
Product interaction	By watching this livestream, I can learn about how the product is used. By watching this livestream, I can learn about the product's technology. By watching this livestream, I can learn about the product's brand market-related information.
Social interaction	While watching this livestream, I will interact with the virtual streamer (for example, following the account, claiming store coupons, and joining the fan group). While watching this livestream, I will communicate with the virtual host about the livestream content. While watching this livestream, I will discuss the content with other viewers.

4.1.2. Details of main study – Case 1

The main study (Case 1) was conducted after the pre-study was completed. The purpose of the main study was to examine the mediating effects of social presence or perceived value. The main study adopted a single-factor (social interaction vs. product interaction) between-subjects design and was conducted on the Credamo platform in December 2023. The survey used a simple random sampling method. Before filling out the questionnaire, respondents were required to answer a screening question: "Have you made a purchase in a virtual streamer's live broadcast room over the past month?" Only those who answered "Yes" qualified as participants for this study. Setting the effect size f to 0.350 and the power to 0.950, the planned sample size was calculated to be 96 or more using G*Power 3.1.9.7 software. A total of 173 questionnaires were collected, of which 73 invalid responses (due to insufficient completion times or failure on lie detection questions) were excluded, leaving 100 valid questionnaires for analysis. The valid sample is characterized by a gender distribution of female = 52 %, male = 48 % and a young-consumer age distribution, with the majority of respondents aged 20 to 40 (78 %). These demographic characteristics closely align with iimedia Research's consumer profile of virtual streamer audiences (iiMedia Research, 2023), which indicates that 89.2 % are aged 22 to 40, with a gender distribution of female = 61.4 %, male = 38.6 %. This alignment suggests that the sample is representative of the actual audience of virtual streamers.

The specific experimental process was as follows: Firstly, participants were randomly allocated to either a product interaction group or a social interaction group. In both groups, the product recommended by the virtual streamer was a watch. Subsequently, subjects viewed a livestream scenario video corresponding to their assigned group. Upon completion of the virtual video, participants were asked to complete scales measuring social presence, perceived value, and purchase intention based solely on the type of virtual streamer interaction observed in the virtual video, without considering product price, brand, or personal preferences.

4.2. Study 2

To test H3 and H4, Study 2 examined the moderating effect of product type (utilitarian vs. hedonic) on the relationship between virtual streamer interaction types (product interaction vs. social interaction) and consumer purchase intention H3H4.

4.2.1. Pre-study

Study 2 employed a 2 (virtual streamer interaction type: product vs. social) × 2 (product type: utilitarian vs. hedonic) between-subjects design. Accordingly, four livestream videos were developed in which a virtual streamer promoted either a utilitarian product (external hard drive) or a hedonic product (perfume), using either product-oriented or social-oriented interaction styles.The specific experimental process was consistent with Study 1. The main difference was that Study 2 employed perfumes and external hard drives as the experimental material, as shown in Fig. 3. 50 participants were recruted to engage in the experiment, who were randomly assigned to either of the two interaction types (product interaction or social interaction). Results of a one-way ANOVA show that the Social Interaction Material condition was rated significantly higher for social interaction compared to the Product Interaction Material condition ($M_{Social\ interaction} = 5.46$ vs. $M_{product\ interaction} = 2.92$, $F(1,48) = 219.23$, $p < 0.001$). On the other hand, participants assigned to the product interaction condition had a significantly greater rate of product interaction than those assigned to the social interaction condition ($M_{Product\ interaction} = 5.64$ vs. $M_{Social\ interaction} = 2.58$, $F(1,48) = 360.14$, $p < 0.001$). The perfumes condition was rated significantly higher for the hedonic product compared to the external hard drives



Utilitarian Product–Product interaction group



Utilitarian Product–Social interaction group



Hedonic Product–Product interaction group



Hedonic Product–Social interaction group

Fig. 3. Video screenshots of Study 2.

condition ($M_{Perfumes} = 5.6$ vs. $M_{Hard\ Drives} = 3.12$, $F(1,48) = 200.17$, $p < 0.001$). Similarly, the external hard drives condition was rated significantly higher for the utilitarian product compared to the perfumes condition ($M_{Hard\ Drives} = 5.30$ vs. $M_{Perfumes} = 2.44$, $F(1,48) = 247.96$, $p < 0.001$). Therefore, intreraction and product types were effectively manipulated.

4.2.2. Details of main study – Case 2

The main study (Case 2) was conducted after the pre-study was completed. The purpose of this main study was to measure the moderating effect of product type (utilitarian vs. hedonic) toward the influence of virtual streamer interaction types (product interaction vs. social interaction) on consumer purchase intention. The study employed the Credamo questionnaire platform for a randomized scenario experiment. Before filling out the questionnaire, respondents were required to answer a screening question: "Have you made a purchase in a virtual streamer's live broadcast room over the past month?" Only those who answered "Yes" qualified as participants for this study. Following the methodology of Faul et al. (2009), we set the effect size f to 0.250 and the power to 0.950; the planned sample size was calculated to be 197 or more using G*Power 3.1.9.7 software. Therefore, the main study recruited 374 participants including 177 males and 197 females, and each participant was randomly assigned to one of the two experimental groups. 173 were excluded after careful screening (mainly due to insufficient completion times or failing the lie detection questions), leaving 201 valid questionnaires for analysis. The valid sample is characterized by a gender distribution of female = 52.2 %, male = 47.8 % and a young-consumer age distribution, with the majority of respondents aged 20 to 40 (85.5 %). These demographic characteristics closely align with iResearch's consumer profile of virtual streamer audiences, which indicates that 89.2 % are aged 22 to 40, with a gender distribution of female = 61.4 %, male = 38.6 %. This alignment suggests that the sample is representative of the actual audience of virtual streamers.

The specific experimental process was as follows: Firstly, participants were randomly divided into four groups. Before the experiment, each group received different instructions designed to prompt their perception of the virtual influencer's interaction type and product type. For product interaction, participants were told that the virtual streamer interacted mainly with the product, brand, and related knowledge. In contrast, for social interaction, they were informed that the interaction centered on communication between the audience, the streamer, and other virtual viewers. For utilitarian products, participants were advised that the recommended products were practical, like external hard drives or toothpaste (with external hard drives selected for this experiment). For hedonic products, they were informed that the recommended products were for enjoyment, such as perfumes. After watching the livestream virtual videos, participants completed scales measuring social presence, perceived value, and purchase intention.

5. Results and discussion

5.1. Result of Study 1

5.1.1. Manipulation check

Given that the experiment implemented two types of interactions, a manipulation check was conducted to verify the operationalization of these interaction types. Results showed a significant difference in perceived value between participants exposed to social interactions with the virtual streamer compared to those in the product interaction scenario ($M_{social} = 5.029$, $SD = 0.388$, $M_{product} = 5.820$, $SD = 0.349$, $F = 67.577$, $p < 0.05$). Additionally, a significant distinction in social presence was observed between the two groups ($M_{social} = 5.9488$, $SD = 0.414$, $M_{product} = 3.968$, $SD = 0.473$, $F = 476.982$, $p < 0.05$). Thus, the manipulation of virtual streamer interaction types was deemed successful.

5.1.2. Variable measurement

To ensure the high reliability and validity of our measurement tools, mature scales from previous literature were adopted to measure the key variables meticulously. Purchase intention was assessed using the live-streaming purchase intention scale developed by Tong et al. (2022). Social presence was measured based on the scale proposed by Lim et al. (2021), while customer-perceived value was primarily derived from scales developed by Jiang et al. (2021), with adjustments made to accommodate the unique characteristics of e-commerce livestreaming. A 7-point Likert scale was employed to evaluate these variables, where 1 represented "strongly disagree" and 7 signified "strongly agree." All scales underwent rigorous reliability and validity tests (overall *Cronbach's Alpha* for the scale was 0.902, with respective *Alphas* of 0.839 for perceived value, 0.890 for social presence, and 0.854 for purchase intention, all exceeding 0.7, passing the reliability test; furthermore, $KMO = 0.904$ and Bartlett's test result was significant with $p < 0.001$, passing the validity test).

5.1.3. Effects analysis

(1) ***Main Effect Analysis.*** Independent sample *t*-tests (as shown in Table 2) revealed significant differences in perceived value and social presence across interaction types. Specifically, product interaction had a more positive impact on perceived value compared to social interaction. Conversely, social interaction was found to have a more substantial positive influence on social presence than product interaction. Furthermore, the intention to purchase was significantly higher for social interaction ($M_{social} = 6.193$, $SD = 0.386$) compared to product interaction ($M_{product} = 5.964$, $SD = 0.371$; $t = 3$, $p = 0.003$). These preliminary findings suggest that social interaction generally outperforms product interaction in stimulating consumer purchase intention, with social interaction particularly effective in enhancing social presence and product interaction more efficacious in boosting perceived value.

(2) ***Mediation Effect Analysis.*** Independent sample *t*-test results revealed that social presence induced by social interactions was significantly more robust than that from product interactions, indicating that social interactions are more effective than product interactions in eliciting social presence in consumers. To further validate the mediating role of social presence, SPSS with PROCESS tools (Model 4) was employed, controlling for gender and age, with 5000 bootstrap resamples to examine the dataset. As shown in Tables 3 and 4, with product interaction as the control group and social interaction as the experimental group, the mediation effect of social presence was significant and fully mediating. Specifically, the type of virtual streamer interaction significantly influenced social presence ($t = 21.64$, $p = 0.000$). Moreover, social presence, acting as the crucial mediator, significantly and positively impacted consumer purchase intention ($t = 3.7072$, $p = 0.0004$). However, no significant direct effect was detected when examining the direct impact of virtual streamer interaction types on consumer purchase intention ($BootLLCI = -0.7575$, $BootULCI = 0.0351$, $p = 0.0424$). This finding strongly suggests that social presence fully mediates the relationship between virtual streamer interaction types and consumer purchase intention, confirming H1.

Similarly, independent samples of the *t*-test results indicated that the perceived value derived from product interactions was significantly higher than that from social interactions, suggesting that product interactions are more effective than social ones in enhancing consumers' perceived value. To further validate the mediating effect of perceived value, SPSS with PROCESS tools (Model 4) was employed, with gender and age as control variables, and the sample data was tested with 5000 bootstrap resamples. As illustrated in Tables 5 and 6, with social

Table 2
Independent sample *t*-test results.

	Type	Number of Cases	Mean (Standard Deviation)	*t*	*p*	Comparison
Perceived value	Watch - Social interaction	43	5.029(0.605)	−7.658	0	Product interaction
	Watch - Product Interaction	57	5.820(0.349)			>Social interaction
Social presence	Watch - Social interaction	43	5.948(0.414)	21.84	0	Social interaction
	Watch - Product Interaction	57	3.968(0.473)			>Product interaction
Purchase intention	Watch - Social interaction	43	6.193(0.386)	3	0.003	Social interaction
	Watch - Product Interaction	57	5.964(0.371)			>Product interaction

Table 3
R-squared and F-values for the mediation effect of social presence.

	Purchase intention		Purchase intention		Social presence	
	t	*p*	*t*	*p*	*t*	*p*
Sex	2.1905	0.0309	1.7806	0.0781	−0.7999	0.4257
Age	0.2785	0.7812	0.3802	0.7046	0.3329	0.7399
Type of interaction	−2.0568	0.0424	3.0077	0.0034	21.6403	0
Social presence	3.7072	0.0004				
R^2	0.2261		0.1141		0.8309	
F	6.9379		4.1222		157.2784	

Table 4
A decomposition table of total effect, direct effect, and mediation effect for social presence.

	Effect	*BootSE*	*BootLLCI*	*BootULCI*	result
Indirect effect	0.5842	0.1826	0.2256	0.9492	Sig.
Direct effect	−0.3558	0.2021	−0.7575	0.0351	–
Total effect	0.2284	0.076	0.081	0.375	Sig.

Table 5
R-squared and F-values for the mediation effect of perceived value.

	Purchase intention		Purchase intention		Perceived value	
	t	*p*	*t*	*p*	*t*	*p*
Sex	1.7108	0.0904	1.7806	0.0781	0.5462	0.5862
Age	−0.1559	0.8764	0.3802	0.7046	1.1309	0.2609
Type of interaction	−5.8305	0	−3.0077	0.0034	8.2471	0
Perceived value	5.0505	0				
R^2	0.3016		0.1141		0.4173	
F	10.2578		4.1222		22.919	

Table 6
A decomposition table of total effect, direct effect, and mediation effect for perceived value.

	Effect	*BootSE*	*BootLLCI*	*BootULCI*	Result
Indirect effect	0.2881	0.0725	0.166	0.4438	Sig.
Direct effect	−0.5166	0.1024	−0.7154	−0.3159	Sig.
Total effect	−0.2284	−0.076	−0.081	−0.375	Sig.

interaction as the control group and product interaction as the experimental group, the mediation effect of perceived value was significant, functioning as a partial mediator. The type of virtual streamer interaction significantly impacted consumers' perceived value ($t = 8.2471$, $p = 0.000$). Meanwhile, as a pivotal factor in consumer purchase decisions, perceived value significantly and positively influenced consumer purchase intention ($t = 5.0505$, $p = 0.000$). Additionally, virtual streamer interaction types' direct and significant effect on consumer purchase intention was observed ($t = -3$, $p = 0.0034$). These findings demonstrate that perceived value partially mediates the relationship between virtual streamer interaction types and consumer purchase intention, confirming H2.

5.2. Result of study 2

5.2.1. Manipulation checks

We designed a 2 (virtual streamer interaction types: product interaction vs. social interaction) × 2 (product types: utilitarian vs. hedonic) between-subjects experiment to investigate these factors' interaction mechanisms.

First, we conducted a manipulation check for the interaction types. Results showed significant differences in perceived value between participants exposed to social interactions ($M = 5.1832$, $SD = 0.362$) and those in product interaction scenarios ($M = 5.96$, $SD = 0.468$; $F = 173.27$, $p < 0.05$). Similarly, significant differences in social presence were observed between the two interaction groups ($M_{social} = 5.998$, $SD = 0.316$, $M_{product} = 5.032$, $SD = 0.286$; $F = 513.362$, $p < 0.05$), confirming the successful manipulation of interaction types.

Second, we performed a manipulation check for the product types. Significant differences in perceived value were found between participants exposed to hedonic products ($M = 5.236$, $SD = 0.417$) and those to utilitarian products ($M = 5.776$, $SD = 0.346$; $F = 12.486$, $p < 0.05$). Likewise, there were significant variations in social presence based on product types ($M_{hedonic} = 5.595$, $SD = 0.523$, $M_{utilitarian} = 5.132$, $SD = 0.475$; $F = 9.286$, $p < 0.05$), validating the successful manipulation of product types.

In summary, the four groups for the experimental materials were successfully manipulated, allowing for the next phase of the experiment.

5.2.2. Variable measurement

To ensure the reliability and validity of measurement tools, mature scales from prior studies were utilized to assess critical variables. The overall *Cronbach's Alpha* for the scale in Study 2 was 0.93, with respective Alphas of 0.875 for perceived value, 0.905 for social presence, and 0.896 for purchase intention, all-surpassing 0.7, passing the reliability test. Furthermore, with $KMO = 0.912$ and a significant Bartlett's test result ($p < 0.001$), the validity test was also passed.

5.2.3. Effects analysis

(1) ***Moderated Mediation Effect.*** Given that both independent and moderating variables were categorical, we employed one-way ANOVA to test for moderation effects. By examining the significance of the interaction term, we could assess whether the

moderation effect was significant. The detailed analysis results are presented in Table 7, showing that when social presence served as the mediator, the interaction item between product and interaction types was significant ($p = 0.007 < 0.05$), indicating a significant moderating effect of product type on the relationship between the interaction type and social presence. Similarly, with perceived value as the mediator, the interaction item was also significant ($p = 0.000 < 0.05$), demonstrating a significant moderating effect of product type on the link between the interaction type and perceived value. These findings confirm a significant moderating role of product type in the influence of virtual streamer interaction types on the mediators—social presence and perceived value.

(2) ***Comparison of Moderation Effects***. According to the results of the general linear model analysis of the interaction effect, a significant interaction was found between the type of virtual streamer interaction and product type ($F = 255.783$, $p < 0.05$), with detailed statistical outcomes depicted in Fig. 4. Further simple effect analyses revealed the following.

When the product type was utilitarian, compared to social interaction, virtual streamer product interaction led to a more significant enhancement in consumer purchase intention ($M_{product\ interaction} = 6.4314$, $SD = 0.3217$, $p < 0.05$; $M_{social\ interaction} = 5.8631$, $SD = 0.3409$, $p < 0.05$).

Conversely, when the product type was hedonic, relative to product interaction, virtual streamer social interaction was more effective in stimulating positive consumer purchase intention ($M_{social\ interaction} = 6.5111$, $SD = 0.2201$, $p < 0.05$; $M_{product\ interaction} = 5.6531$, $SD = 0.2802$, $p < 0.05$).

These findings highlight the distinct roles of interaction types based on product categorization, emphasizing how virtual streamer strategies should be tailored to product types to foster consumer purchase intention optimally.

(3) ***Social Presence and Perceived Value Interaction Tests***. Initially, the interaction effect of virtual streamer interaction types and product types on perceived value was examined. ANOVA results indicated a significant interaction ($F = 21.065$, $p < 0.05$), with detailed statistics illustrated in Fig. 5. Specifically, when the product was utilitarian, compared to social interaction, product interaction was found to elicit a higher level of perceived value among consumers, yielding the highest mean value across all groups ($M_{product\ interaction} = 6.2108$, $SD = 0.4073$; $M_{social\ interaction} = 5.1920$, $SD = 0.4316$; $F = 25.179$, $p < 0.05$).

Subsequently, the interaction effect of virtual streamer interaction types and product types on social presence was investigated. The ANOVA results also revealed a significant interaction ($F = 7.473$, $p < 0.05$), with specific statistical outcomes presented in Fig. 6. Notably, when the product was hedonic, social interaction, as opposed to product interaction, generated a heightened sense of social presence among consumers, reaching the highest

Table 7
Significance tests for moderation effects.

Equation 1: Moderating Effect of the Moderator on the Relationship Between the Independent Variable and the Mediator (Social Presence)			
Predictor Variables	*F*	*p*	R^2
Product type	22.705	0	0.758
Type of interaction	601.281	0	
Product type * Type of interaction	7.473	0.007	
Equation 2: Moderating Effect of the Moderator on the Relationship Between the Independent Variable and the Mediator (Perceived Value)			
Predictor Variables	*F*	*p*	R^2
Product type	24.581	0	0.566
Type of interaction	207.828	0	
Product type * Type of interaction	21.065	0	
Equation 3: Moderating Effect of the Moderator on the Relationship Between the Independent Variable and the Dependent Variable			
Predictor Variables	*F*	*p*	R^2
Product type	3.711	0.056	0.572
Type of interaction	8.465	0.004	
Product type * Type of interaction	255.783	0	

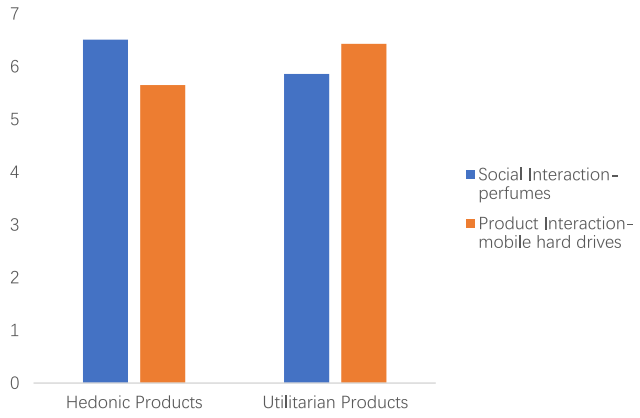


Fig. 4. Interaction effect of virtual streamer interaction types and product.

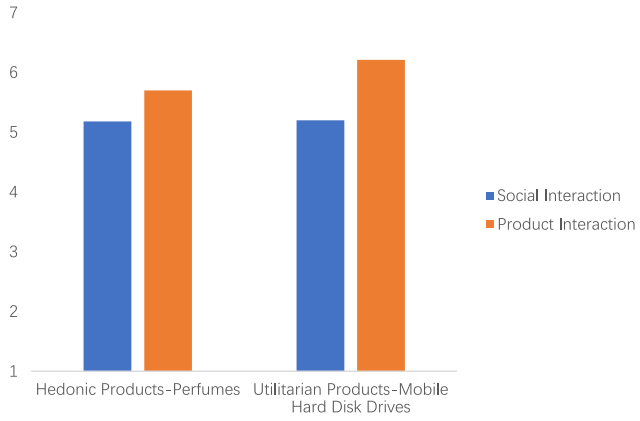


Fig. 5. Interaction effect of virtual streamer interaction type and product type on consumers' perceived value.

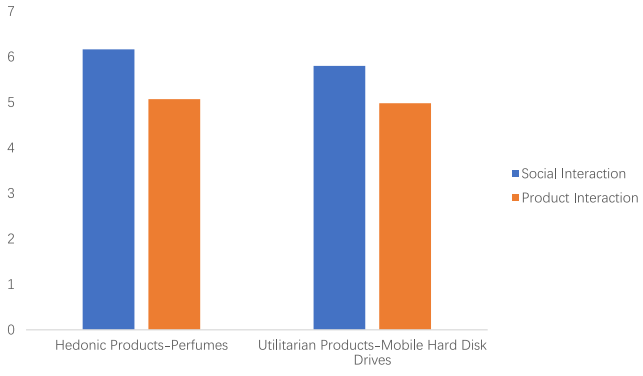


Fig. 6. Interaction effect of virtual streamer interaction type and product type on consumers' sense of social presence.

mean across the four groups ($M_{social\ interaction} = 6.164$, $SD = 0.27399$; $M_{product\ interaction} = 5.0735$, $SD = 0.2789$; $F = 21.825$, $p < 0.05$). These findings confirm H3 and H4.

In summary, the product type exerts a regulatory effect on the relationship between virtual streamer interaction types and both social presence and perceived value. Specifically, when promoting utilitarian products, virtual streamers' product interaction (vs. social interaction) is more effective in positively influencing consumer purchase intention through the mediating role of perceived value. Conversely, when promoting hedonic products, social interaction (vs. product interaction) proves more efficacious in enhancing consumer purchase intention virtually via the mediation of social presence.

5.3. Discussion

Previous research has primarily explored the interactivity of virtual streamers and the mediating role of social presence on the relationship between interactivity and consumer purchase intention. Building on this foundation, our study differentiates interactions into product-focused and social categories. It further introduces product type as a moderating variable to examine the mediating effects of social presence and perceived value. The main findings of our research can be summarized as follows.

(1) Both virtual streamer interactions—social and product-focused—positively stimulate consumer purchase intention through the mediating mechanisms of perceived value and social presence. These findings suggest that virtual streamer engagements leverage social presence and perceived value regardless of the interaction type to facilitate consumer purchasing decisions effectively. Our study further illuminates those social interactions enhance social presence, whereas product-focused interactions are more proficient in amplifying perceived value, thereby triggering stronger purchase intentions through their respective strengths.
(2) Product type plays a pivotal moderating role in this process. The effects of social and product-focused interactions vary significantly depending on the product category being promoted. Practically, when utilitarian products are endorsed, product-focused interaction by virtual streamers has a more pronounced positive impact on consumer purchase intention through the perceived value pathway. Conversely, when hedonic products are promoted, social interaction through social presence exerts a more prominent positive influence on consumer purchase intention. This insight implies that utilitarian product marketers should prioritize enhancing virtual streamers' product-focused interaction capabilities, whereas for hedonic products, optimizing social engagement and anthropomorphism of virtual streamers becomes crucial to better cater to consumer psychological needs and elevate purchase intention.

5.3.1. Theoretical contributions

The present study makes three salient contributions to the literature on consumer behavior and e-commerce, particularly in the context of virtual streamers in live commerce platforms. These advancements offer fresh perspectives on applying social presence and perceived value theories in digital interactive environments, enrich our understanding of the strategic use of virtual streamers, and provide actionable insights for practitioners. The integrated framework emphasizes the importance of social presence in co-creation, arguing that organizations should prioritize improving social cues and human features in their service delivery to create deeper consumer engagement and collaborative value generation.

First, this study contributes to a clearer understanding of how product and social interactions function differently in virtual streamer-consumer engagement. While prior research has examined perceived value and emotional connection as outcomes of virtual streamer interactions, it has not distinguished the specific roles played by different interaction types. This lack of differentiation limits theoretical clarity regarding how virtual streamers influence consumer behavior through distinct engagement strategies. By conceptually linking product interactions with perceived value theory and social interactions with social presence theory, this study provides a more structured explanation of the psychological mechanisms underlying consumer responses. In doing so, it extends the applicability of both theories to digitally mediated, socially immersive contexts such as livestreaming commerce, and enriches their explanatory depth by highlighting how interaction types shape consumer perceptions and decision-making.

Second, our study integrates the SDL framework proposed by Vargo and Lusch (2004) to synthesize the effects of product and social interactions, overcoming the limitations of single-path explanations. Doing so presents a comprehensive virtual viewpoint that accounts for both the functional benefits emphasized by product interactions and the emotional connections fostered through social interactions (Yao et al., 2024). This holistic approach, grounded in SDL, offers a nuanced understanding of how virtual streamers engage with consumers, enhancing the explanatory power of existing theories and offering a more complete picture of the mechanisms underlying consumer decision-making in livestreaming environments.

Lastly, introducing product type as a moderator variable represents a significant step forward in refining theoretical frameworks. This study bridges the gap between theory and practice by demonstrating that the effects of virtual streamer interactions differ based on whether the product is utilitarian or hedonic, rendering the theoretical model more closely aligned with real-world complexities. This nuanced understanding of product categorization guides strategic decisions regarding the optimal use of virtual streamers for different product categories, enhancing the practical utility of the research findings (Hu and Ma, 2023).

In summary, this study's theoretical implications refine and expand upon established theories and provide a more dynamic and practical lens to view virtually and leverage virtual streamers in e-commerce strategies. Integrating social presence, perceived value, and product type considerations within the SDL framework offers a robust theoretical foundation for future research and practical applications in the increasingly digitalized retail landscape.

5.3.2. Managerial implications

Based on the findings of this study, there are several actionable strategies that e-commerce businesses and marketers can implement to enhance the effectiveness of virtual streamers in boosting consumer purchase intention. Practically, this study provides valuable insights into customer relationship management, showing that firms should engage in technology and procedures that improve social presence to establish better, more loyal customer connections. By leveraging the understanding of the differential impacts of interaction types (product interaction vs. social interaction) and product categories (utilitarian vs. hedonic), brands can tailor their virtual streamer strategies to maximize consumer engagement and sales conversion. Businesses may create more successful customer engagement strategies emphasizing on fostering emotional and relational ties by comprehending the relationship between social presence and value perception. Here are some practical implications derived from the research:

Customize Interaction Strategies According to Product Type. Recognizing that product-focused interactions enhance perceived value for utilitarian products more effectively, brands should emphasize detailed product information, specifications, and usage demonstrations during livestreams for items like electronics or household appliances. Conversely, for hedonic products, such as fashion or entertainment items, social interactions should be prioritized to foster social presence

and emotional connections. This might involve more storytelling, audience engagement, and shared experiences.

Optimize Social Presence for Hedonic Products. To stimulate stronger purchase intentions for hedonic goods, virtual streamers should create a virtual vibrant and engaging social atmosphere, perhaps through interactive games, live chat competitions, or collaborations with real users to share their experiences. This aligns with findings that social interactions significantly increase social presence and purchase intention for such products.

Enhance Product Knowledge Delivery for Utilitarian Products. Virtual streamers should be equipped with comprehensive product knowledge when promoting utilitarian products, ready to answer queries promptly and informatively. Implementing Q&A sessions or trivia quizzes related to the product can enhance the perceived value and satisfy consumers' curiosity.

6. Conclusion, limitations, and future research directions

The study illuminated the mediating roles of social presence and perceived value toward virtual streamer interactions' influence on consumer purchase intent, but not without limitations. Primarily, the young consumer-oriented sample narrows generalizability, highlighting a need for broader demographic inclusion. Additionally, concentrating on just two product categories—mobile hard drives and perfumes—underscores the necessity for a wider product scope in future research. Further, unexamined consumer characteristics suggest a gap in understanding individual response variability to virtual interactions.

To address these limitations, three future research avenues are proposed: (1) longitudinal studies to assess the longevity of effects amid technological and market evolution; (2) cross-cultural explorations to decipher how cultural subtleties influence attitudes toward virtual streamers; and (3) examining personalization tactics, including adaptable appearances and communication styles, to refine targeted marketing and deepen engagement understanding. These paths will enhance our comprehension of virtual streamers' e-commerce role and optimize strategy development.

CRediT authorship contribution statement

Hao Liu: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Peilin Zhang:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **Hongqing Cheng:** Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Najmul Hasan:** Writing – review & editing, Visualization, Validation, Methodology, Investigation. **Raymond Chiong:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization.

Disclosure statement

No potential conflict of interest was reported by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Measurement scales

Constructs	Items	References
Product interaction	I can learn about the usage information of the product through watching this part of the livestream.	Yuan and Huang (2022)
	Through watching this section of the livestream, I can gain insights into the brand and market-related information of the product.	
	I can learn about the usage information of the product through watching this part of the livestream.	
Social interaction	I will respond to the virtual influencer's interactions during this segment of the livestream.	
	I will engage in conversations with the virtual streamer about the livestream content while watching this part.	
	I will engage in discussions with fellow viewers about the livestreaming content or have discussions while watching this part of the livestream.	
Perceived value	Purchasing this practical/entertainment product makes me feel that I am getting my money's worth.	Jiang et al. (2021)
	Acquiring this practical/entertainment product aligns with my needs.	
	Buying this practical/entertainment product would have positive effects on the environment and society.	
	Purchasing this practical/entertainment product makes me feel relaxed and delighted.	
Social presence	During this livestream marketing event, I experience a sense of social connection.	Lim et al. (2021)
	During this livestream marketing, I can sense a human warmth and enthusiasm.	
	Within this livestream marketing, I feel a context of interacting with others.	
	In this livestreaming marketing event, I am able to perceive the presence of other involved individuals.	
Purchase intention	After viewing this livestreaming video, I became interested in making a purchase.	Tong et al. (2022)
	After viewing this livestreaming video, I'm willing to purchase the product being presented.	
	After viewing this livestreaming video, I would consider purchasing the presented product.	
	After viewing this livestreaming video, I will likely buy the product being presented.	

Data availability

Data will be made available on request.

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